

Acute Pericarditis — Clinical Reference for APP Educators

Teaching reference · Department of Medical Education · 2026

1. Overview and Etiology

Acute pericarditis is inflammation of the pericardium, the fibroserous sac surrounding the heart. It is the most common cause of acute chest pain in adults under 40 presenting to emergency and urgent care settings, and it is a high-yield teaching diagnosis because it is frequently mistaken for acute coronary syndrome (ACS).

In previously healthy young adults, the majority of cases are **idiopathic or viral ("idiopathic-viral")** — presumed viral with no pathogen identified. Less common but important causes include: recent myocardial infarction (early post-MI pericarditis, days 1–4, vs. Dressler syndrome, weeks later), uremia, autoimmune disease (systemic lupus erythematosus, rheumatoid arthritis), tuberculosis, malignancy, radiation, and trauma. In the outpatient and urgent care setting, post-viral idiopathic pericarditis dominates: a viral upper respiratory infection 1–2 weeks before chest pain onset is a classic history.

2. Diagnostic Criteria

The clinical diagnosis requires **at least 2 of 4** validated criteria:

1. **Typical pericardial chest pain** — sharp and pleuritic, worse with inspiration and when supine, improved by sitting forward and leaning; may radiate to the trapezius ridge (nearly pathognomonic when present)
2. **Pericardial friction rub** — a scratchy, high-pitched, "leather creaking" sound, best heard at the left lower sternal border with the patient sitting forward at end-expiration; may be evanescent (comes and goes)
3. **Typical ECG changes** — diffuse concave ST-segment elevation with PR-segment depression (see the accompanying ECG teaching note)
4. **New or worsening pericardial effusion** on echocardiogram

Supportive findings include an elevated C-reactive protein (CRP) or erythrocyte sedimentation rate (ESR), fever, and leukocytosis.

TEACHING POINT

Because no single finding is universally present, trainees who anchor on "the ECG looks like a STEMI" or "there is no rub" will miss the diagnosis. The criteria deliberately reward recognizing the *pattern*: young patient + pleuritic/positional pain + diffuse concave ST elevation = pericarditis until proven otherwise.

3. Physical Examination

The examination is often more informative than trainees expect. Have the patient sit forward and lean slightly; auscultate at the left lower sternal border at end-expiration. A pericardial friction rub — when present — has been described as having up to three components (atrial systole, ventricular systole, early diastole), though a single-component rub is common. The rub may disappear as effusion develops (the fluid separates the inflamed layers).

Assess for tamponade physiology in every suspected case: hypotension, elevated jugular venous pressure, muffled or distant heart sounds (Beck triad), and pulsus paradoxus greater than 10 mmHg. Kussmaul sign (JVD rising with inspiration) suggests constrictive physiology rather than pericarditis.

4. Electrocardiogram — What to Expect

The ECG evolves through four stages (detailed in the accompanying ECG teaching note):

Stage	Timing	Findings
I (acute)	Days	Diffuse concave ST elevation, PR-segment depression; reciprocal changes limited to aVR (ST depression + PR elevation)
II	1–3 weeks	ST segments normalize; T waves flatten
III	Weeks	T-wave inversion (may mimic ischemic T waves)
IV	Weeks–months	T waves normalize

PR-segment depression is the most specific early finding and is seen in up to 80% of cases in stage I. A normal ECG does not exclude pericarditis.

5. Laboratory and Imaging Workup

Recommended initial evaluation for suspected acute pericarditis:

- **Inflammatory markers:** CRP (also guides prognosis and the decision to taper therapy — normalize before tapering) and ESR
- **Troponin:** to screen for myopericarditis. Elevated troponin with pericarditis = myopericarditis; these patients warrant cardiology involvement, echo, and activity restriction
- **Complete blood count, BMP (renal function — exclude uremic pericarditis), TSH**
- **Echocardiogram for every patient** — to define the presence, size, and hemodynamic significance of any pericardial effusion
- Chest X-ray (may show an enlarged cardiac silhouette with large effusion; screen for pneumonia, mass)

Routine viral serologies are not recommended — they do not change management.

Tuberculous and malignant pericarditis should be considered when the presentation is subacute, recurrent, or accompanied by weight loss, night sweats, or a large hemorrhagic effusion.

6. Treatment — First-Line Therapy

The goals of therapy are pain relief, resolution of inflammation, and prevention of recurrence. First-line treatment is **combination therapy: a high-dose NSAID plus colchicine**.

- **NSAID:** ibuprofen 600–800 mg three times daily (or aspirin 750–1000 mg every 8 hours), continued until symptom resolution, then tapered over 2–4 weeks
- **Colchicine:** 0.5 mg twice daily if ≥ 70 kg (once daily if under 70 kg), continued for **3 months**. Colchicine roughly halves the recurrence rate and is the single most evidence-supported adjunct in pericarditis care

- **Gastric protection** with a proton-pump inhibitor while on NSAIDs
- **Activity restriction** (see Section 7)

Avoid corticosteroids as first-line therapy except when NSAIDs/colchicine are contraindicated (aspirin allergy, anticoagulation, renal failure) or the etiology is autoimmune, uremic, or pregnancy-related. Corticosteroids are associated with a *higher recurrence rate* — a counterintuitive teaching point — and, when required, should be the lowest effective dose tapered over weeks.

TEACHING POINT

Trainees commonly reach for prednisone because "inflammation equals steroids." Pericarditis is the canonical exception: steroid use is an independent predictor of recurrences, and colchicine is the adjunct that changes long-term outcomes.

7. Return to Activity and Sports

Athletes and active patients must refrain from competitive sports and strenuous exercise until: (1) chest pain has resolved, (2) CRP has normalized, (3) ECG changes have stabilized, and (4) repeat echocardiogram shows resolution of any effusion. A **minimum restriction period of 3 months** is standard for athletes. Non-athletes may return to light activity as symptoms resolve. Recurrence on resumption of activity should prompt re-evaluation before progressive return.

8. Red Flags — When to Escalate

ESCALATE IMMEDIATELY IF

- Any sign of tamponade: hypotension, tachycardia, JVD, muffled heart sounds, pulsus paradoxus >10 mmHg — obtain urgent echocardiogram; pericardiocentesis if hemodynamically compromised
- Large or enlarging pericardial effusion, or effusion with diastolic collapse on echo
- Elevated troponin (myopericarditis), reduced ejection fraction, arrhythmias, or heart failure signs
- Fever >38°C persisting beyond 5–7 days, immunosuppression, recent cardiac surgery, trauma, or anticoagulation (hemopericardium risk)

- Recurrent pericarditis (occurs in roughly 15–30% of first episodes; colchicine reduces this)

9. Teaching Points for APP Fellows

1. The diagnosis is a pattern, not a single test — two of four criteria make the diagnosis.
2. The most feared mimic is STEMI; the most common mimic is musculoskeletal pain. Positional, pleuritic quality + trapezius radiation should trigger an ECG.
3. Diffuse concave ST elevation + PR depression with aVR changes = pericarditis pattern, not a coronary territory pattern.
4. Treat with NSAID + colchicine, not steroids; restrict activity; normalize CRP before tapering.
5. Always look for tamponade physiology — the diagnosis is common, but the complication is what kills.

Prepared as teaching source material. Consistent with current cardiology consensus guidance on acute and recurrent pericarditis (ESC/Adler et al. framework).